Speaker Information

[Name] Ke-Hsin Chen

[Institute] Brain Science Center, National Defense Medical University

[Position] Assistant professor

[Email Address] chenkehsin@gmail.com

[Lab / Personal Website] N.A.

Talk Information

A multi-contrast MRI brain template for Taiwanese macaque

Non-human primates (NHPs) have long been critical models in biomedical research. Compared to other lab animals, NHPs are phylogenetically closer to humans, and thus provide better models of the health and diseases in terms of genetics, anatomy, physiology and behavior. NHPs' large brain, high intelligence and sociability make them especially suitable for the studies of higher cognitive functions and neuropsychiatric disorders. Taiwanese macaque (*Macaca cyclopis*) is the native primate living in Taiwan and is a close relative of the rhesus (*Macaca mulatta*) and Japanese macaques (*Macaca fuscata*). However, the feasibility of using it in biomedical research, especially in neuroscience, has rarely been studied. To facilitate this species to be used in brain researches, a standard anatomical template is required for data analysis and comparison across subjects and studies. As a first step, we build an in-vivo MRI brain template with multiple MR contrasts and an averaged skull model from CT images. These basic image tools can be used for surgical planning, parameter simulation for neuromodulation tools and so on. A comprehensive brain atlas will be developed in the future by combining high-resolution ex-vivo MRI image and histological images via whole-brain sectioning.

Research Overview (for collaboration discussion)

[Research Interests] experience-induced whole brain connectivity alteration, caused by neuromodulation, learning, or drug addiction etc.

[Current Research Projects] Comparative neuroanatomy and histology of Taiwanese macaque and other non-human primates

[Techniques] MRI, tape-transfer cryosection

[Aspects or techniques of interest for future collaboration] DNA-related process and analysis, any type of AI-aid image processing for histological images, animal body motion, or emotional facial expression

Short Bio

a mother of a three-year-old kid